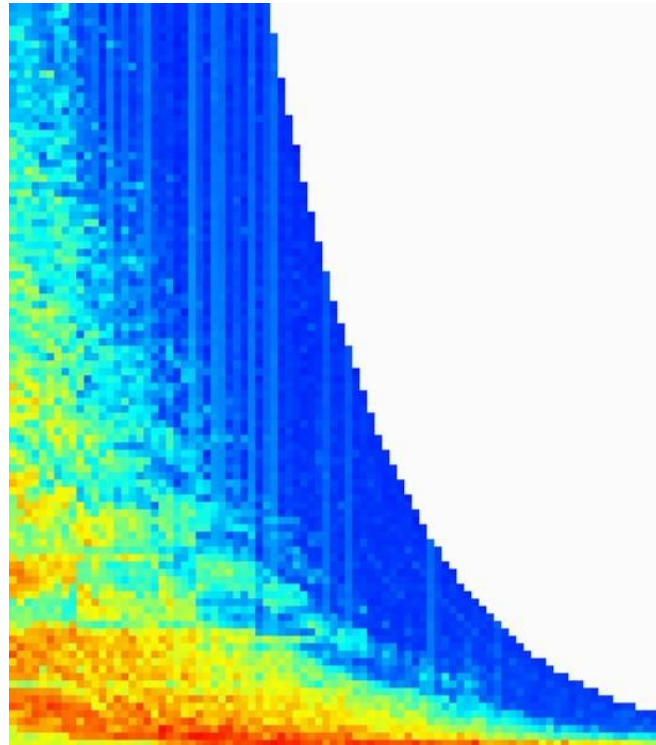




SYDE252 - W01B - Signal Basics

Dr. Matt Borland

What are we doing today:

- Formally define a signal using math.
- Define Continuous and Discrete Signals.
- Introduce the concepts of Energy and Power in reference to signals.
- Talk about manipulating signals.

You need to be able to math!

- Expectation

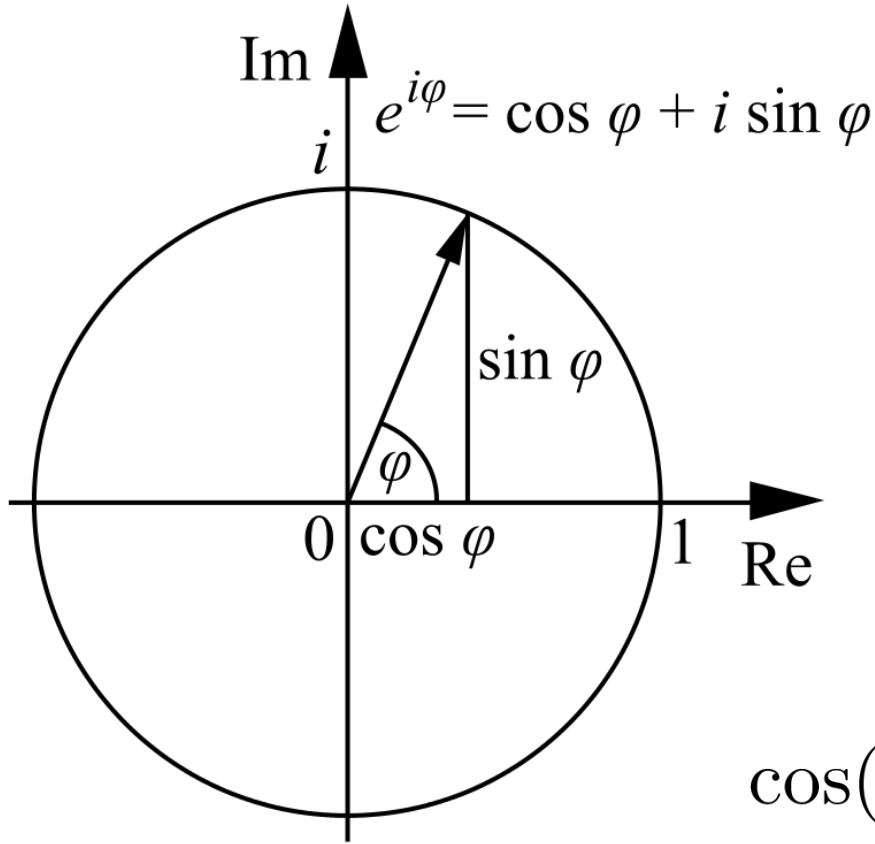
- You should be able to do simple integration;

$$\int_a^b x^2 dx, \int te^t dt$$

- You should know and apply basic trigonometry identities

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$$

Make sure to go through Chapter B -
Background, and fill in any gaps!



Euler's Formula!

$$\cos(x) = \operatorname{Re}(e^{ix}) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sin(x) = \operatorname{Im}(e^{ix}) = \frac{e^{ix} - e^{-ix}}{2i}$$

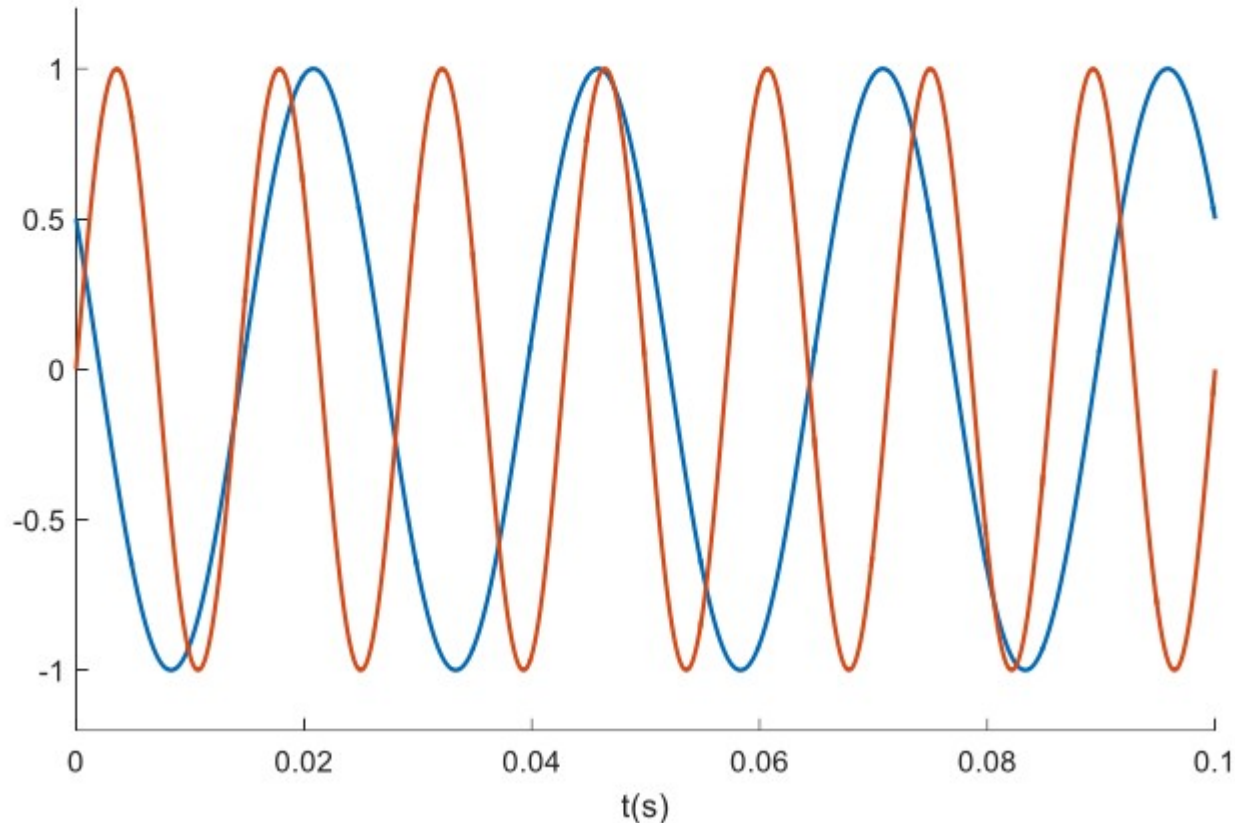
https://en.wikipedia.org/wiki/Euler%27s_formula

What is a Signal?

Mathematically, what's a signal?

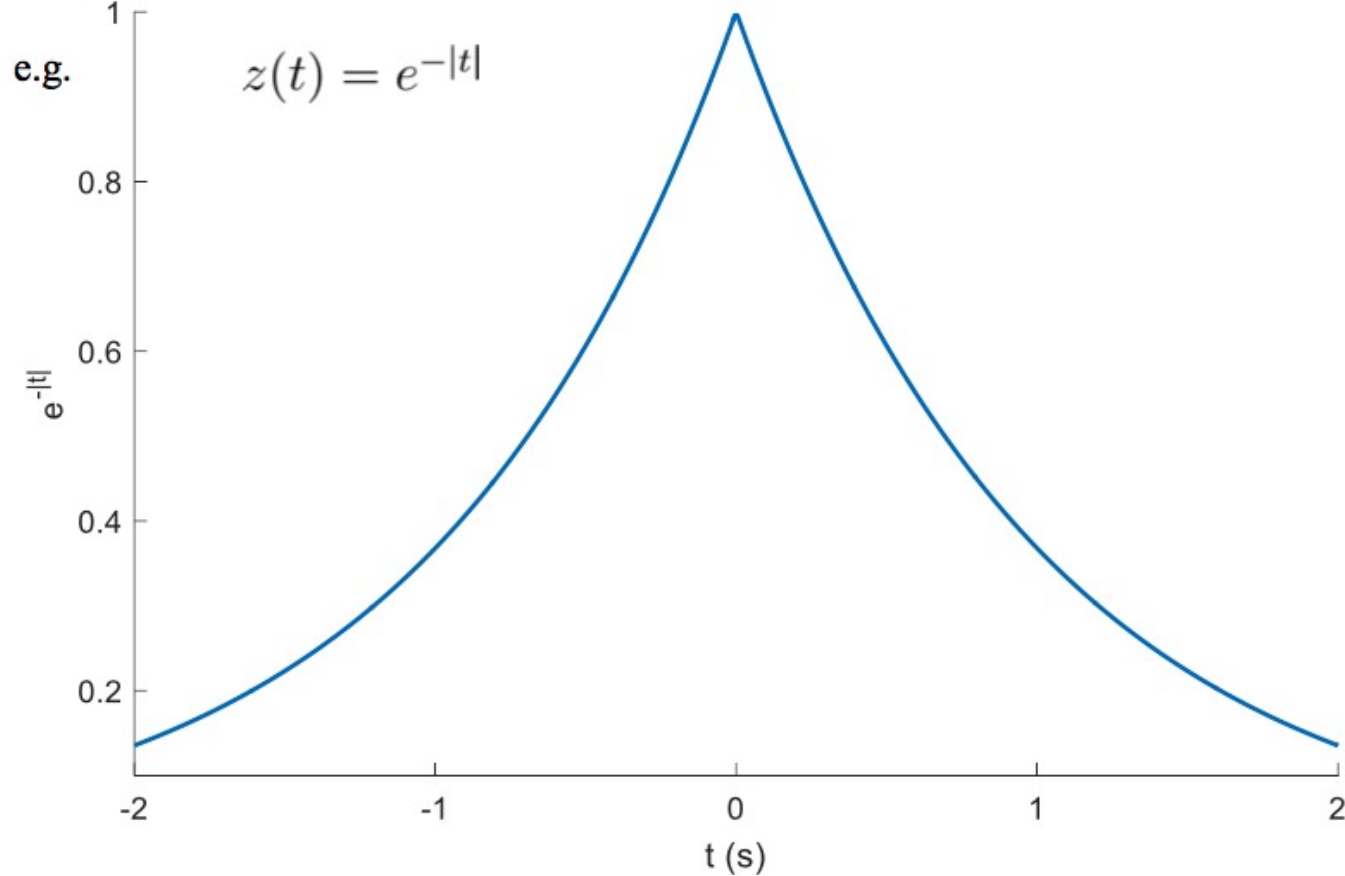
A variable that change its value with respect to the value of an independent variable

e.g. $x(t) = \cos(2\pi ft + \theta) = \cos(\omega t + \theta)$



Mathematically, what's a signal?

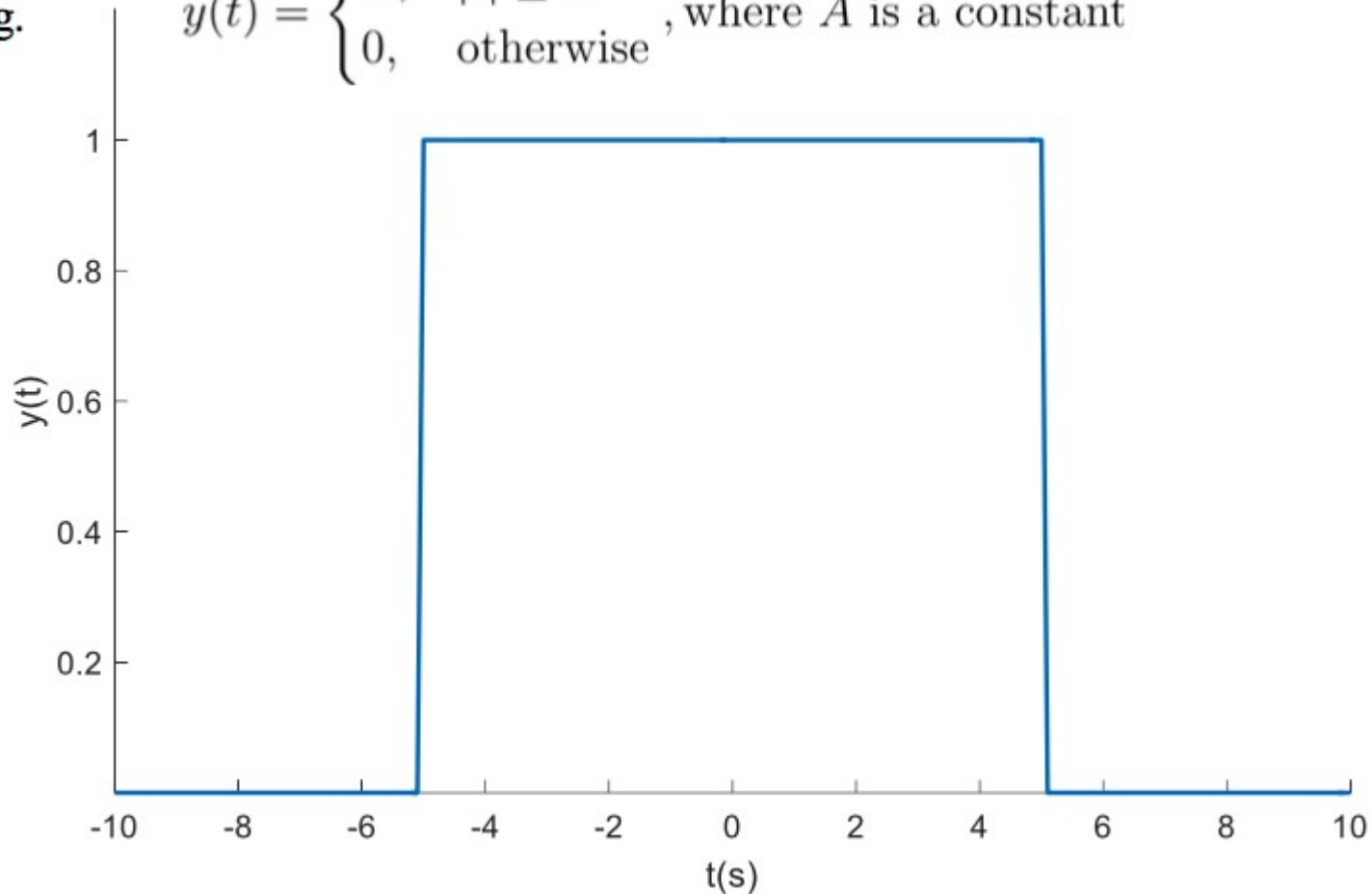
A variable that change its value with respect to the value of an independent variable



Mathematically, what's a signal?

A variable that change its value with respect to the value of an independent variable

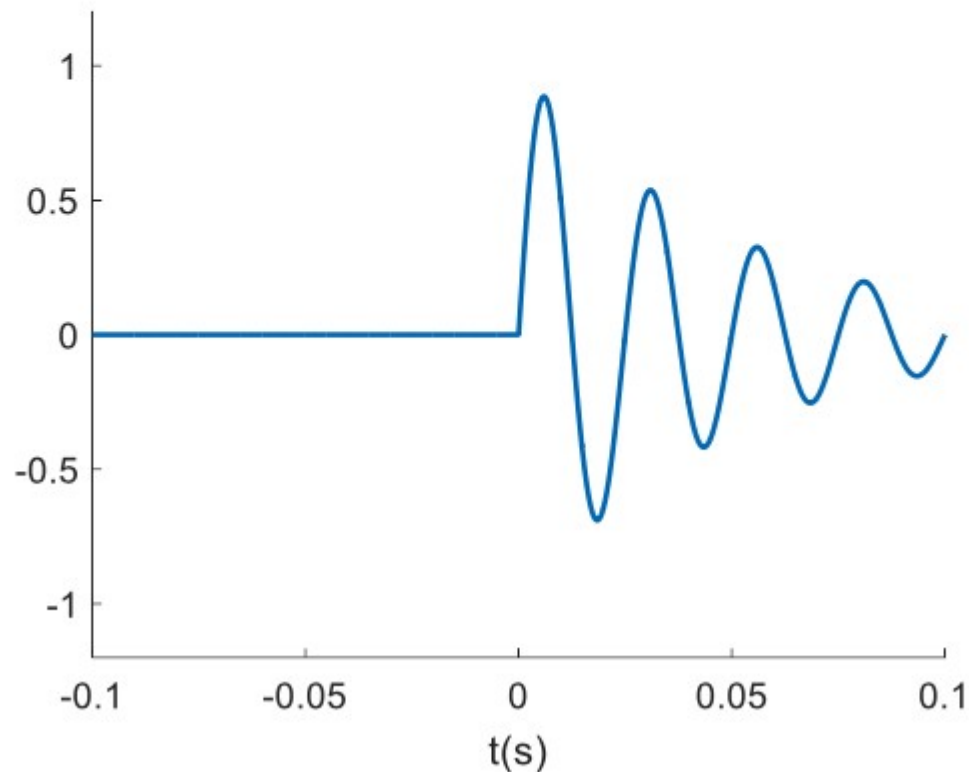
e.g. $y(t) = \begin{cases} A, & |t| \leq 5 \\ 0, & \text{otherwise} \end{cases}$, where A is a constant



Mathematically, what's a signal?

A variable that change its value with respect to the value of an independent variable

e.g.
$$g(t) = \begin{cases} \cos(80\pi t) * e^{-20t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

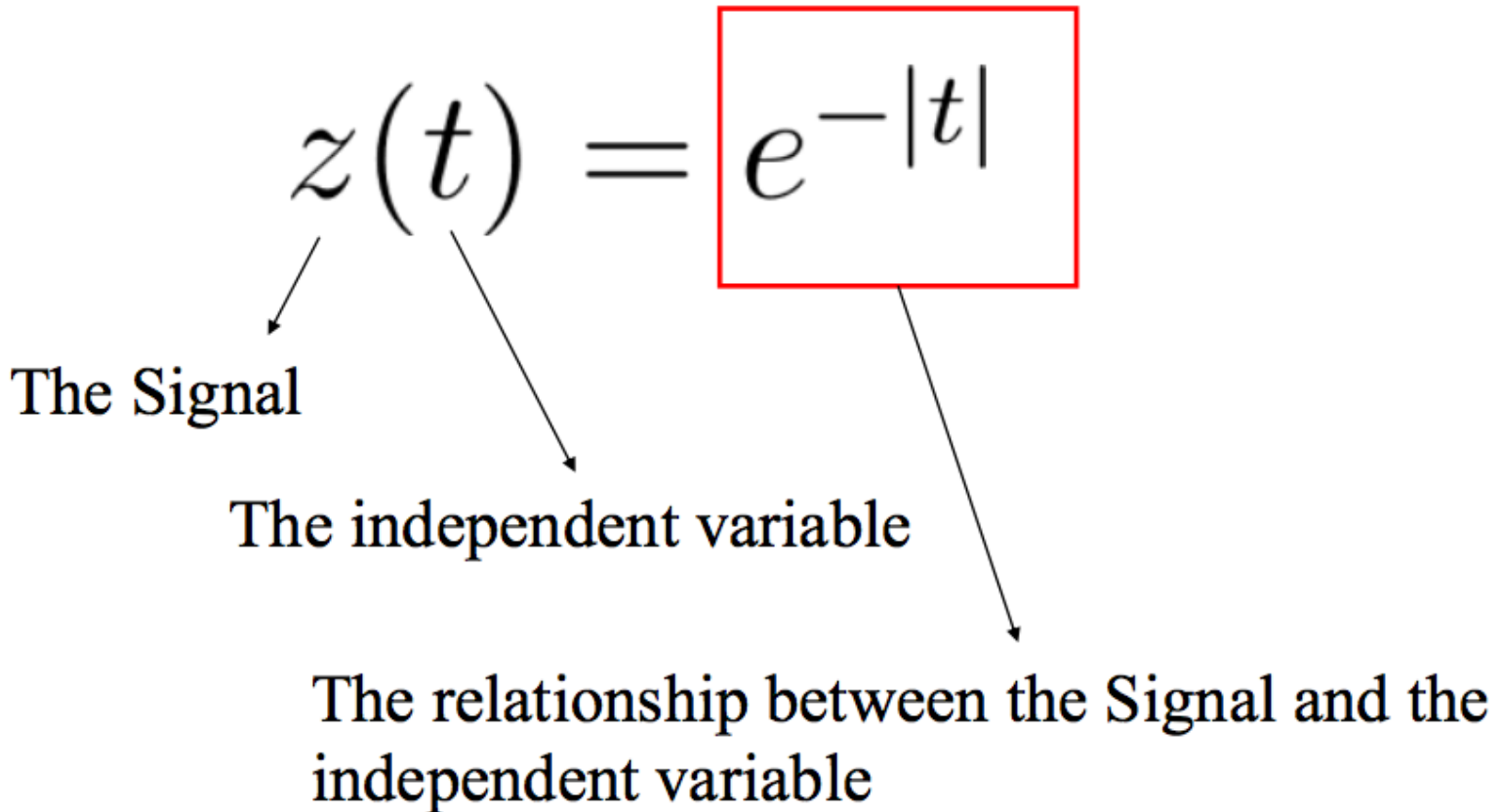


Continuous and Discrete Signals

Continuous Time (CT) Signals

A variable that change its value with respect to the value of another variable

Continuous-time signal

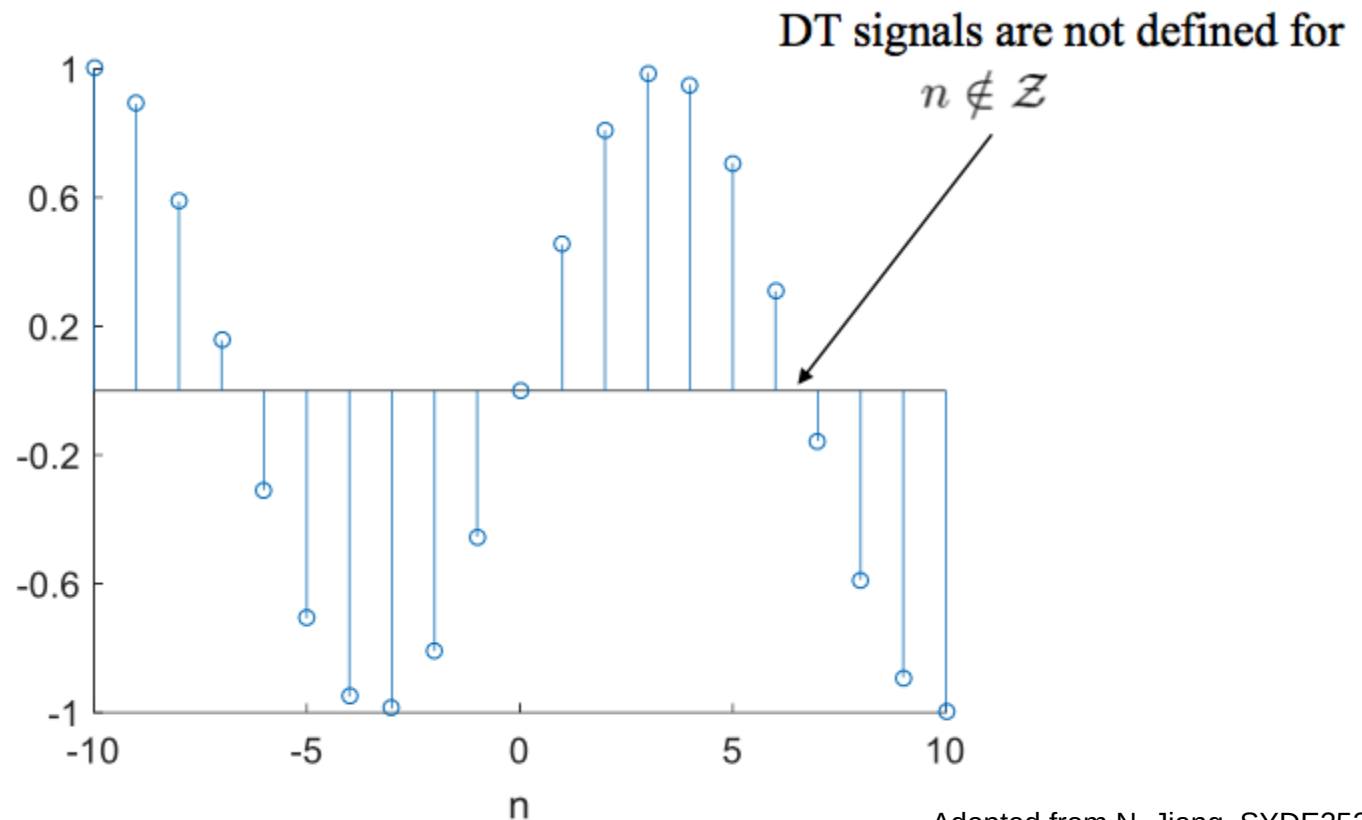


Discrete Time (DT) Signals

Different from CT signal, the independent variable of a different type of signal can only take integer values: $x[n], n \in \mathbb{Z}$. This type of signal is called the **Discrete-time Signal**, or in short, **DT signal**

e.g.

$$x[n] = \sin\left(\frac{3}{20}\pi n\right)$$



Discrete Time (DT) Signals

A variable that change its value with respect to the value of another variable

Discrete-time signal

$$z[n] = \sin\left(\frac{3}{40}\pi n\right)$$

The Signal

The independent variable

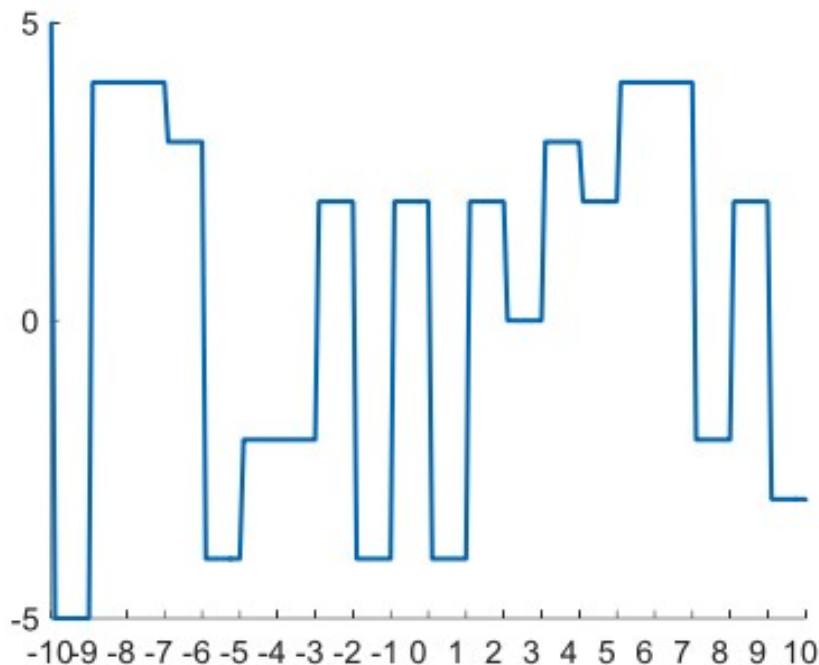
The relationship between the Signal and the independent variable

CT + DT Signals

Definition of Signal

Continuous-time Signal $x(t), y(t), z(t), g(t)$, where $t \in \mathcal{R}$

Discrete-time Signal $x[n], y[n], z[n], g[n]$, where $n \in \mathcal{Z}$



Is the signal on the left a CT signal or a DT signal?

Measuring Signals

Measuring (Characterizing) Signals

How do we define the size of a signal (over time)?

e.g. The voltage across a resistor of $R \Omega$ is $v(t)$, the total **energy** over t_1 and t_2 is:

$$\int_{t_1}^{t_2} p(t) dt = \frac{1}{R} \int_{t_1}^{t_2} v^2(t) dt \quad (\text{J})$$

And the average **power** over the time interval of t_1 and t_2 is:

$$\frac{1}{t_2 - t_1} \int_{t_1}^{t_2} p(t) dt = \frac{1}{R(t_2 - t_1)} \int_{t_1}^{t_2} v^2(t) dt \quad (\text{W})$$

The (total) **energy** of an arbitrary CT signal, $x(t)$, over the time interval of t_1 and t_2 is:

$$E_{t_1, t_2} = \int_{t_1}^{t_2} |x(t)|^2 dt$$

The (average) **power** of an arbitrary CT signals, $x(t)$ over the time interval of t_1 and t_2 is:

$$\begin{aligned} P_{t_1, t_2} &= \frac{1}{t_2 - t_1} E_{t_1, t_2} \\ &= \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt \end{aligned}$$

Power and Energy of CT Signals

The total energy of an arbitrary CT signal, $x(t)$, over the time interval $t = -\infty \rightarrow \infty$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

When the amplitude of $x(t)$ does not go to 0 as $|t|$ goes to infinity this won't converge and the signal energy is infinite.

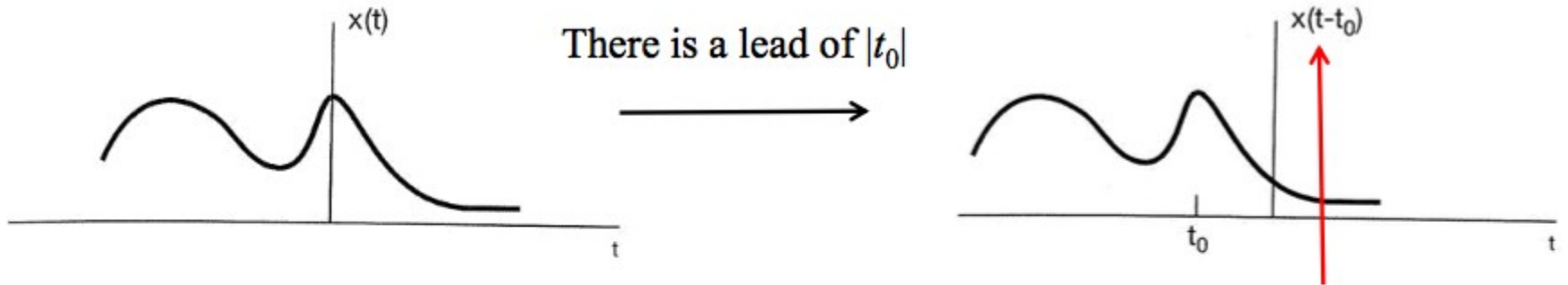
The power of an arbitrary CT signal, $x(t)$, over the time interval $t = -\infty \rightarrow \infty$

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

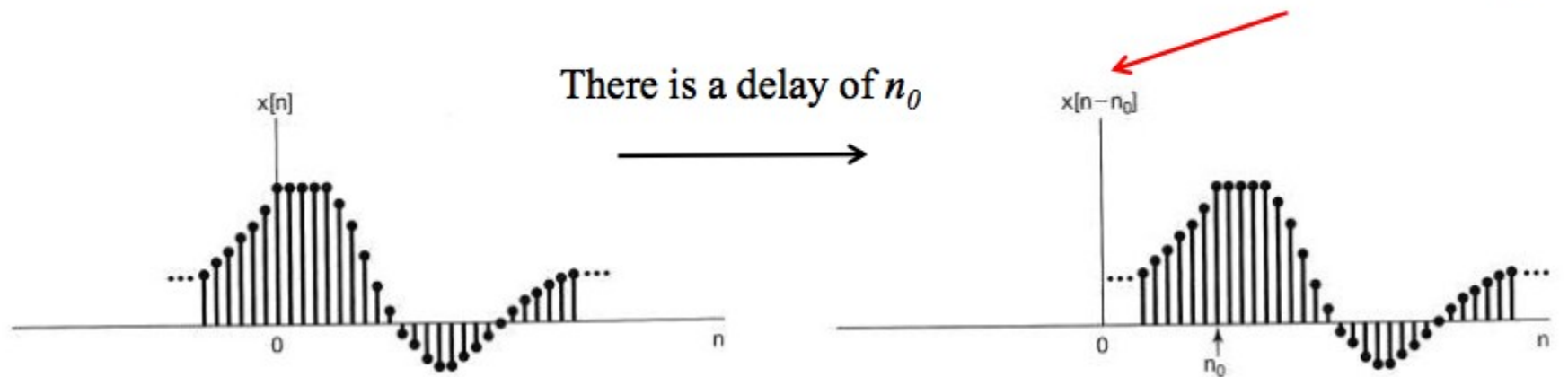
$$x_{rms} = \sqrt{P_x}$$

Manipulating Signals

Time Shifts

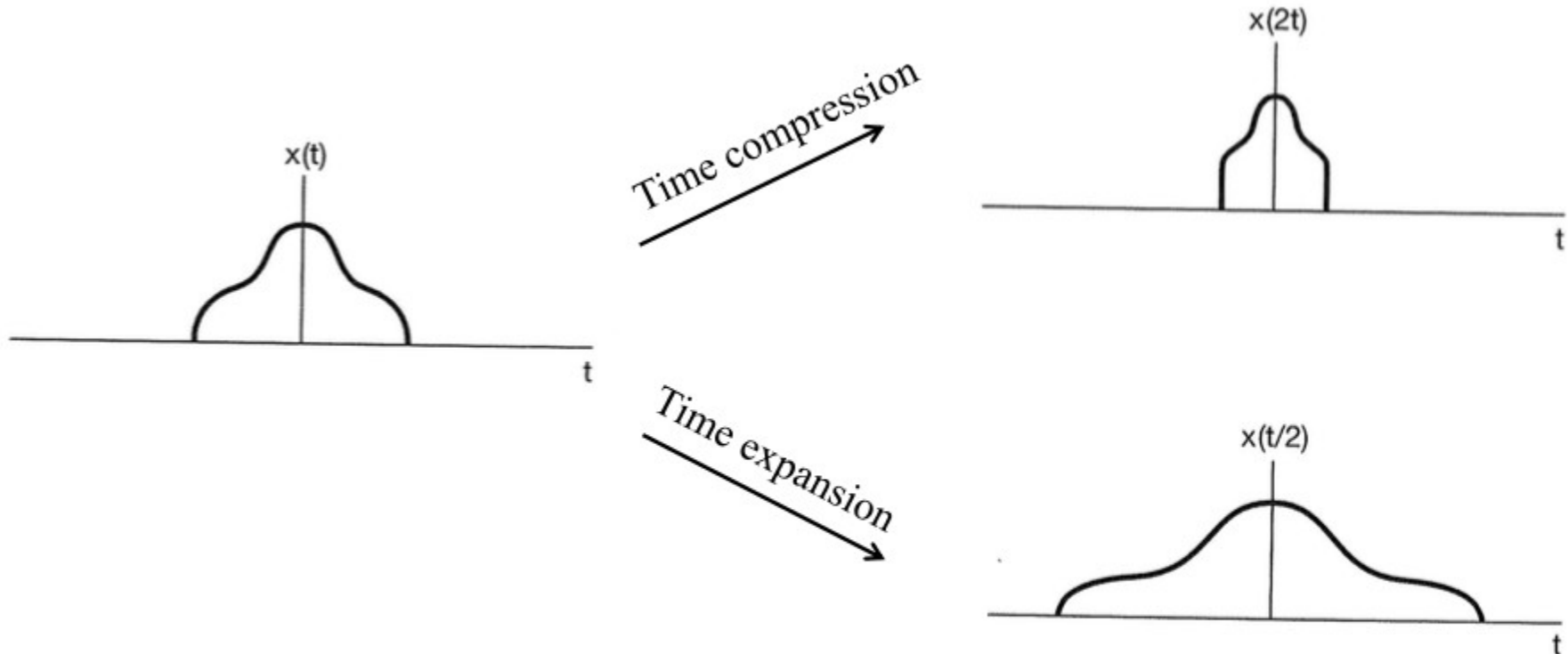


The direction of the shift is **NOT** just determined by the sign



t_0 and n_0 are just constants and can be negative numbers.

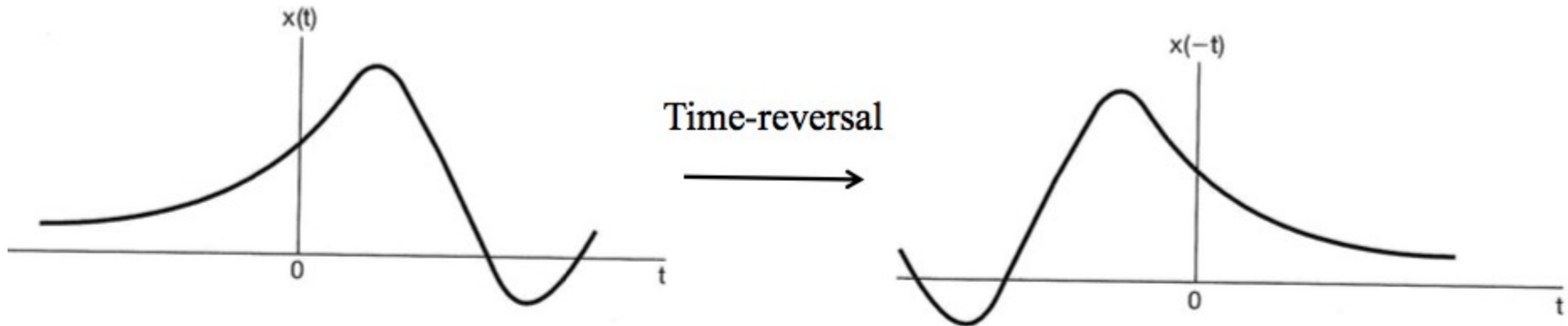
Time Scaling



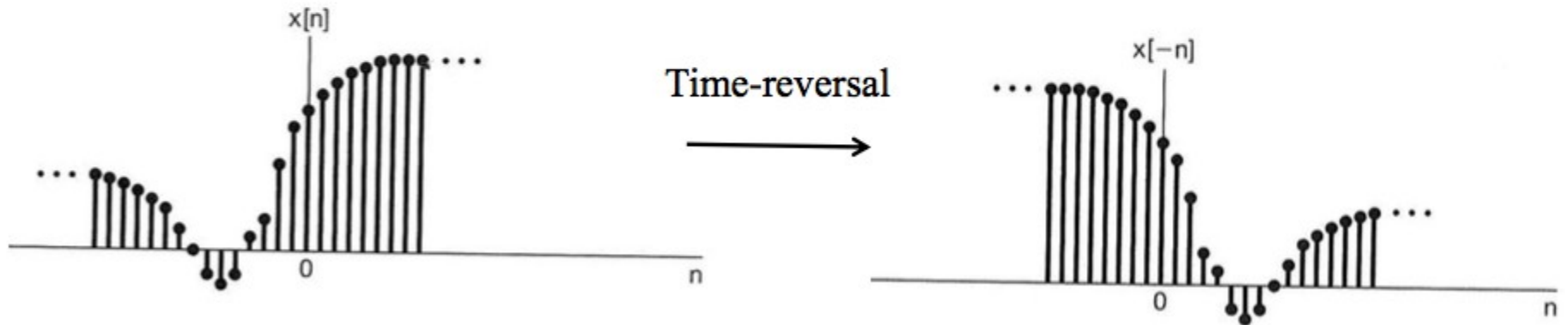
When $a > 1$, $x(at)$ is a **time compression** of $x(t)$ by a factor of a ;

When $1 > a > 0$, $x(at)$ is a **time expansion** of $x(t)$ by a factor of $1/a$;

Time Reversal



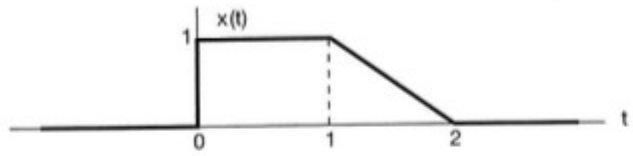
$x(-t)$ is the **time-reversal** of $x(t)$



$x[-n]$ is the **time-reversal** of $x[n]$

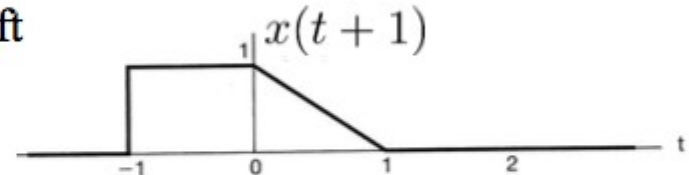
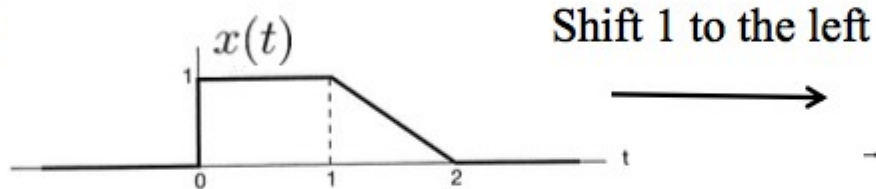
Combining those all together at once:

Example: Given $x(t)$, draw $x(-\frac{3}{2}t + 1)$

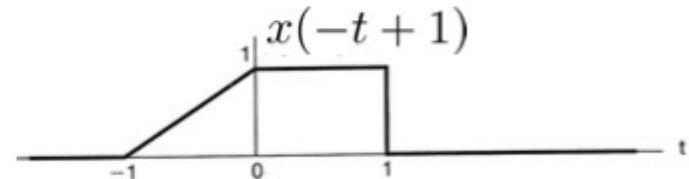
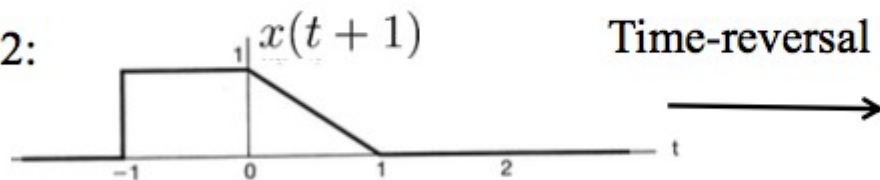


Starting from the transformation **furthest** away from the I.V., and work your way towards the I.V., step by step

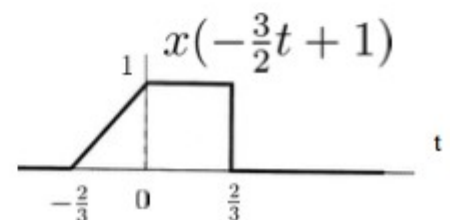
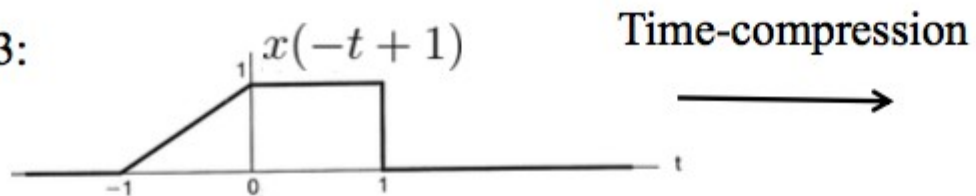
Step 1:



Step 2:



Step 3:



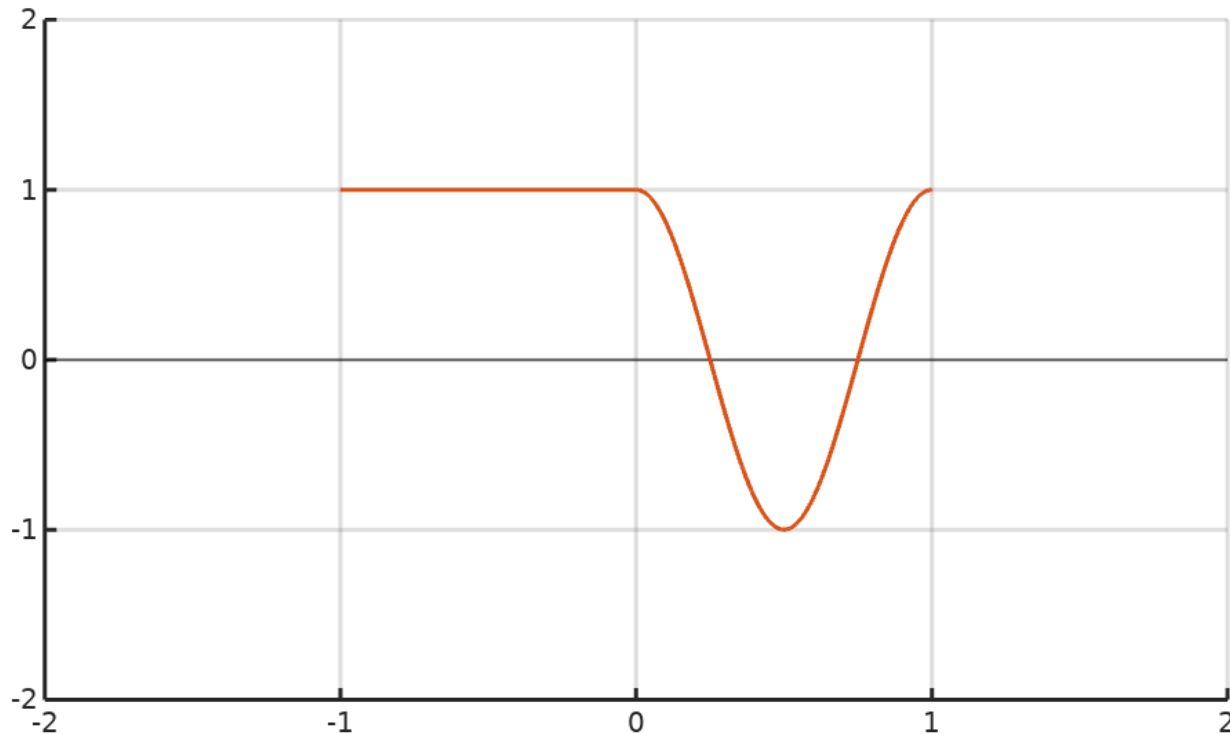
Shift first, then scale and reverse!

Try it out:

$$-1 < t < 0 : u(t) = 1$$

$$0 \leq t < 1 : u(t) = \cos(2\pi t)$$

Draw: $u(\frac{1}{2}t - 1)$



What did you learn today:

- The time domain representation of a signal is related to its frequency domain representation.
- Continuous Signal and Discrete Signals are different.
- We can “measure” signals, or characterize their properties using math.
- Once we characterize the signals we can use that to do things that are useful.
- We can manipulate signals.